

QI RECEIVER 5V (BQ51013B) + 20mA CHARGER (MCP73832)

BATTERY + 3.3V LDO + VBAT MONITOR

nRF52832 CHIP-DOWN + 32MHz XTAL + RF FRONT-END + SWD

ANALOG FRONT-END: 2-OP-AMP POTENTIOSTAT (+0.512V)

DISCRETE POTENTIOSTAT – wireless–charged 3-electrode chronoamperometry

E_{cell} = WE – RE = 1.024V – 0.512V = +0.512V fixed. I_{WE} = (V(ADC_P)–V(ADC_N))/Rf

DESIGN NOTES / VERIFY BEFORE LAYOUT

- CHARGING: two-stage per TI guidance for small cells. BQ51013B = Qi receiver with regulated 5V (VSOUT). ILIM = RILIM = 250/IMAX = R1+R2 = 2.52k -> IMAX 100mA (HW limit 120mA). MCP73832-2 (4.2V) charges the LIR2032: ICHG = 1000/RPROG = 1000/49.9k = 20mA (0.5C). termination ~7.5% = 1.5mA, open-drain STAT -> CHG_N (100k pullup to 3V3).
- RESONANT TANK: L1 11uH (TDK WR202010-18M6-ID, 20x20x1.0mm, DCR 350mR). C1 = 1/(L*(2pi*100kHz)^2) = 230n -> 220n COG (fs 102kHz). C2 -> fd ~1MHz -> 2.2n COG. Retune both whenever the coil changes. Small coil = tighter TX alignment; fine at our 0.13W.
- CHIP-DOWN RF: copy Nordic's QFN48 reference layout verbatim (trace geometry included). FL1 (Johanson 2450FM07A0029) replaces the discrete LC match for nRF52 single-ended ANT. R15/C34 = antenna tee tuning slots (fit OR / DNP initially, tune on real board with VNA). E1 = 0402 micro antenna: keep the datasheet ground keepout exactly; expect finer tuning. DEC2, DEC3 unconnected per QFN48 reference; no 32k crystal -> firmware uses LFRC synth.
- TS/CTRL: 10k NTC against the LIR2032 holder cuts wireless power if the cell heats (plain 10k = sense defeated, still a valid safe strap). EN1=EN2=GND: wireless enabled.
- APE_PWR is driven by nRF GPIO P0.06 set to high-drive. Load = REF35+OPA2391 ~ 60uA. Power the AFE off between runs to avoid electrode polarization.
- WE net: guard ring driven from VREF, no soldermask opening at TIA input, clean flux.
- RF/CF options: 1M/470p (default, +2.1uA/-0.9uA FS), 5.1M/100p (427/181nA), 10M/47p (217/92nA). rLohm + oversample are runtime settings in firmware: recalibrate after a swap.
- R10/C21 control-amp compensation are DNP; fit only if the CE-RE loop oscillates.
- LIR2032 has no protection circuit: firmware must System-OFF below 3.0V (VBAT_DIV, AIN2).
- SAAD: differential P=Ain1(ADC_P) N=Ain0(ADC_N), gain 1/2..4, oversample 64x default.
- XL1/XL2 (P0.00/P0.01) free: no 32.768k crystal fitted. NFC pins unused -> plain GPIO/NC.

FILE INFORMATION:

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Title: Discrete Potentiostat – wireless 3-electrode chronoamperometry

Size: A2 Date: 2026-07-10 Rev: A

KiCad E.D.A. kicad-clit 7.0.11+dfsg-1build4 Id: 1/1

DESIGN NOTES / VERIFY BEFORE LAYOUT

- 1. CHARGING: two-stage per TI guidance for small cells. BQ51013B = Qi receiver with regulated 5V (VSOUT), ILIM: RILIM = 250/IMAX = R1+R2 = 2.52k -> IMAX (HW limit 120mA). C1=220-2.2uF, C2=(1.2-2.2)uF, charging current: ICHG = ICHG_MAX = 100/0.9V = 100mA (0.5C). termination -75% = 1.5uH, open-drain STA1 = CHG1 (100k pullup to 3V3).
- This replaces the earlier BQ51050B direct-charge (TI not recommended <20mA).
- 2. RESONANT TANK: L1 11uH (TDK WR2020110-18M8 -10, 20x20x1.0mm, DCR 350mR). C1 = 1/(L*(2pi*100kHz)^2) = 230n -> 220n COG (f5 102kHz). C2 => fd -1MHz -> 2.2n COG. R1 and R2 both are 100k, 1% tolerance, Small coil = tighter TX alignment; fine at our 0.15W.
- 3. CHIP-POWER: copy Nordic's QFN48 reference layout verbatim (trace geometry included).
- R11 (Johanson 2450FM07A0029) replaces the discrete LC match for nRF52 single-ended ANT. FL15/C34 = antenna tee tuning slots (fit OR / DNP initially, tune on real board with VNA).
- E1 = 0402 micro antenna: keep the datasheet ground keepout exactly; expect finer tuning. D1 = 0402 3.3V 32k pullup to 3.3V crystal pullup. C32 crystal pullup to 3.3V crystal synth.
- 4. TS/CTRL: 10k NTC against the LIR2032 holder cuts wireless power if the cell heats (pinning 10k = sense defeated, still a valid safe strap). EN1=EN2=DND: wireless enabled.
- 5. AFE_PWR is driven by nRF GPIO P0.06 set to high-drive. Load = REF35+OPA2391 - 60uA. Lower the CE off to help with low electrode polarization.
- 6. WIRELESS GUARD: 100k/100pF (217V/32pF) 100M/47pF (217V/32pF) 100M/47pF (217V/32pF).
- 7. RF/CF options: 1M/470p (default, +2.1uA/-0.9uA FS), 5.1M/100p (427/181nA), 10M/47p (92nA). rf_omn + oversample are runtime settings in firmware; recalibrate after a swap.
- 8. R04/C21 control-amp compensation are DNP; fit only if the CE-RE loop oscillates.
- 9. X1 has no protection circuitry (no ESD protection, no ESD System-100, DIV1, DIV2).
- 10. SADC: differential P=AIN1(AFC_P) - N=AIN0(AFC_N) gain 1/2.4, oversample 64x default.
- 11. XL1/XL2 (P0.00/P0.01) add, no 32.768k crystal fitted. NFC pins unused -> pin GPIO/NC.